STD PROJECT | DASHBOARD DOCUMENTATION

Spotify Music Trends Dashboard

100 Years of Sound, 1922 to 2021

An interactive Tableau dashboard built on 586,600 Spotify tracks

586,600	98,504	27.6	4.4%	3.83
Total Tracks	Total Artists	Avg Popularity	Explicit %	Avg Duration (min)

PROJECT DETAILS

Project	STD Project
Topic	Evolution of popular music on Spotify, 1922 to 2021
Dataset	Spotify Dataset 1921–2020, 600k+ Tracks (Kaggle)
Tool	Tableau Public / Desktop
Dashboard	5 KPIs | 3 Filters | 5 Charts
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Date	October 2026

Contents: Problem statement and solution, dataset, KPIs, filters, flowchart, 5-point chart interpretations, conclusion.

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How to read this document

The report follows the STD project brief: dataset overview and link, KPI and filter purpose, a flowchart of the dashboard, and the standard 5-point interpretation (What do you see, Positive scenario, Negative scenario, Conclusion, Suggestion) for every chart.

1. Problem Statement and Solution

1.1 Background

Spotify holds one of the largest music catalogues in the world. Every track is described by audio features (energy, danceability, acousticness, valence and more), a release year, a popularity score and an explicit-content flag. Together these columns record how the sound, content and appeal of music have changed over a century, but only in a raw table of more than half a million rows.

1.2 Problem statement

Problem
A catalogue of 586,600 tracks across 100 years cannot be read as raw rows. Artists, labels, streaming platforms and analysts cannot quickly see **how much music is being released, how its sound and content have shifted, who is most popular today, and which musical traits go with popularity**. Without a clear visual summary, decisions about releases, playlists, content policy and marketing rest on guesswork.

1.3 Objectives

- Turn the raw Spotify data into a clean, analysis-ready dataset.
- Summarize the whole catalogue in five headline KPIs that can be read in seconds.
- Answer five business questions with five charts, each explained in the standard 5-point format.
- Let the viewer explore any decade, explicit setting or content type through interactive filters.
- Document the work with a flowchart that shows how the dashboard is read.

1.4 Proposed solution

Solution
An interactive **Tableau dashboard** (1400 x 900) built on a cleaned version of the dataset. It combines **5 KPI cards**, **5 charts** and **3 filters** on a single page. The charts form one story: how much music exists, how it sounds, how its content changed, who leads today, and what makes a hit. Filters re-slice every KPI and chart together, so the same page can answer questions for any era.

1.5 Business questions answered

#	Question	Answered by
1	How has the size of the music catalogue grown over time?	Graph 1: Tracks released per year
2	How has the sound of music changed across decades?	Graph 2: Audio features by decade
3	Has explicit content increased?	Graph 3: Explicit % by decade
4	Who are the most popular artists today?	Graph 4: Top 10 artists by average popularity
5	Do more danceable songs perform better?	Graph 5: Popularity by danceability band

1.6 Intended users

Artists and producers (what sound is working), **labels and marketers** (who and what to back), **streaming platforms** (content controls and playlisting) and **students or analysts** (a worked example of dataset-to-dashboard).

2. Dataset Overview and Link

2.1 Topic and dataset

The dashboard tells the story of how popular music changed over 100 years (1922 to 2021) using **586,600 tracks** from Spotify's catalogue. Spotify measures every track with audio features. Combined with release year, popularity and explicit flags, these show how music moved from the acoustic recordings of the 1920s to today's streaming hits.

2.2 Dataset link and source

> **Source**
> Public Kaggle dataset **Spotify Dataset 1921-2020, 600k+ Tracks** by Yamac Eren Ay, collected through the Spotify Web API.
> **Link:** https://www.kaggle.com/datasets/yamaerenay/spotify-dataset-19212020-600k-tracks
> **Tool used:** Tableau (Public / Desktop).

2.3 Files in the dataset

File	Rows	Key columns
tracks.csv	586,672	Track name, artists, popularity (0 to 100), duration, explicit flag, release date, 11 audio features
artists.csv	1,162,095	Artist name, followers, genres, artist popularity
dict_artists.json	573,856 artists	Up to 20 related artists per artist (not used in this dashboard)

2.4 Key audio features in plain language

Feature	Scale	Meaning
Acousticness	0 to 1	How acoustic a track sounds (1 = fully acoustic, no electronic production)
Energy	0 to 1	Intensity and activity: fast, loud and noisy tracks score high
Danceability	0 to 1	How suitable the track is for dancing (rhythm stability, beat strength, tempo)
Valence	0 to 1	Musical positiveness: high = cheerful, low = sad or tense
Popularity	0 to 100	Spotify score based on recent play counts (favours recent music)
Speechiness	0 to 1	Amount of spoken words; above 0.66 is treated as Spoken Word in this project

All audio features range from 0 to 1, except loudness (decibels) and tempo (beats per minute). The data runs up to 16 April 2021, so **2021 is a partial year**.

3. Data Preparation

The raw files were cleaned into three Tableau-ready tables before the dashboard was built: **tracks** (586,600 rows), **track-artists** (757,098 rows) and **artists** (81,776 rows).

#	Problem in original data	Action taken
1	71 tracks had no name	Removed
2	1 track dated 1900 (outlier; real data starts 1922)	Removed
3	Release date mixed full dates, year-only and year-month formats	Extracted a clean Year column
4	No way to group by era	Added Decade column (1920s to 2020s)
5	Duration in milliseconds	Converted to Duration (min)
6	Explicit stored as 0/1	Changed to Yes / No
7	22,598 audiobooks, poems and podcasts mixed with songs	Added Content Type: Spoken Word if speechiness > 0.66, else Music
8	Danceability is a 0 to 1 decimal	Added Danceability Band: Low (0 to 0.4), Medium (0.4 to 0.7), High (0.7 to 1.0)
9	Artists stored as list text, e.g. ['Justin Bieber', 'Giveon']	Converted to plain text
10	18% of tracks have multiple artists	Built a track-artist table with one row per track-artist pair
11	Artists file held 1.16M artists, most with no tracks	Kept the 81,776 artists that appear in tracks
12	Genres stored as list text; 74% empty	Converted to plain text; empty marked Unknown
13	Unneeded columns (key, mode, time signature, raw date)	Dropped; remaining columns renamed for readability

Limitation
About 60,000 tracks are re-releases or remasters of the same song. They are separate releases on Spotify, so they were kept. Counts of "tracks" therefore mean **releases**, not unique songs.

4. The Dashboard

The finished Tableau dashboard (1400 x 900) shows **five KPIs, five charts and three filters** on one page. The figure below shows the default view with all filters set to All.

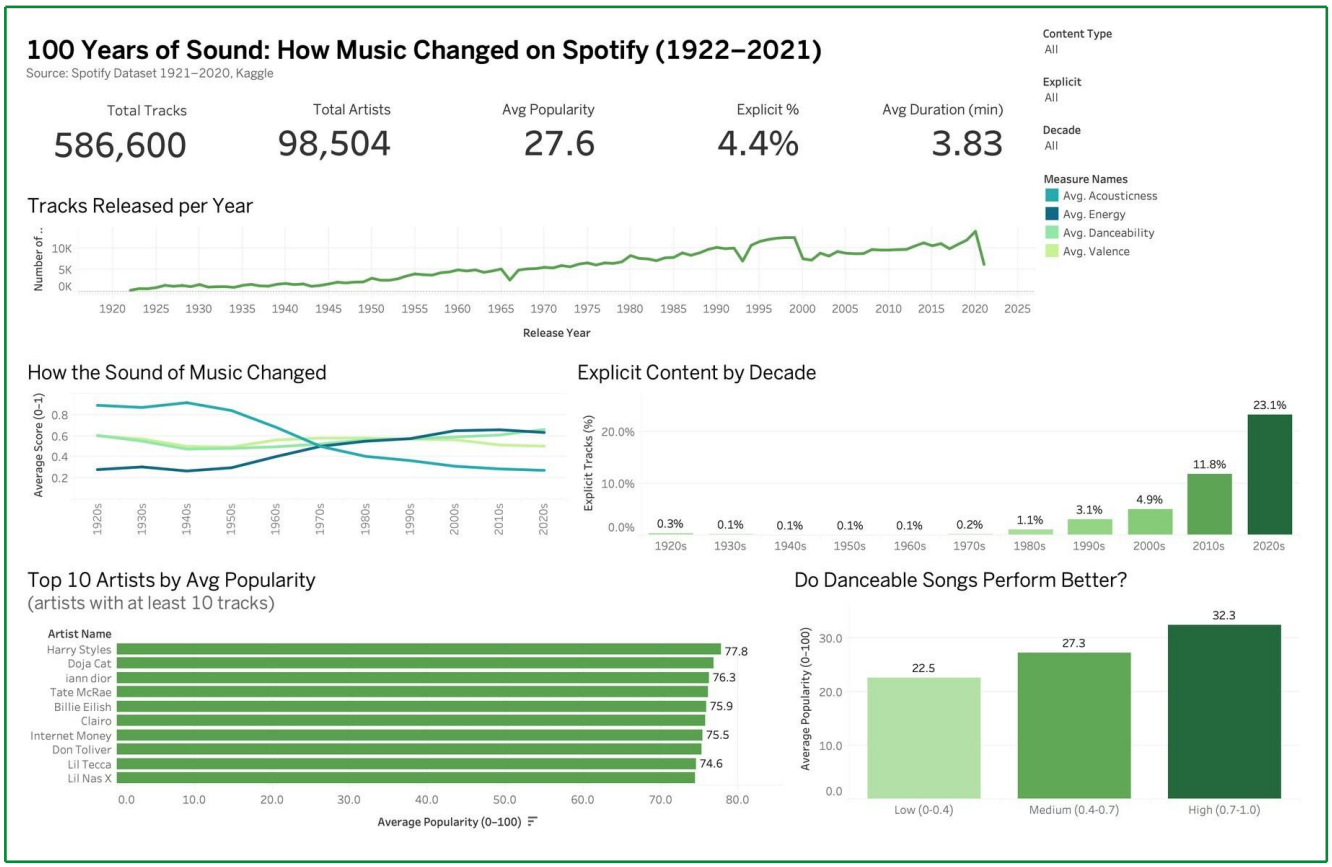


Figure 1: Final dashboard, all filters set to All

Layout at a glance

Zone	Contents	Purpose
Top band	Title and source line	Tells the viewer the topic and data source
KPI row	5 KPI cards	Headline numbers before any chart
Chart area	Graph 1 (full width); Graphs 2 and 3; Graphs 4 and 5	The five-step story
Right panel	Content Type, Explicit and Decade filters	Interactive control of every KPI and chart

5. KPIs and Why They Are Needed

Five KPI cards sit at the top of the dashboard so the viewer gets the headline numbers before reading any chart.

KPI	Value (all data)	Need / purpose	Tableau calculation
Total Tracks	586,600	Shows the size of the catalogue analysed and the scale behind every chart	`COUNTD([Track ID])`
Total Artists	98,504	Shows how many artists contribute, i.e. how diverse the catalogue is	`COUNTD([Artist ID])`
Avg Popularity	27.6 / 100	Benchmark of overall listener interest; each decade or artist can be compared with it	`AVG([Popularity])`
Explicit %	4.4%	Share of songs with explicit lyrics; a content and audience-suitability indicator	`SUM(IIF([Explicit]="Yes",1,0)) / COUNT([Track ID])`
Avg Duration	3.83 min	Typical song length; a hint of listening habits and track format	`AVG([Duration (min)])`

Why KPIs are needed

- They **summarise 586,600 rows into five numbers** the viewer can read in seconds.
- They **give context to every chart** below them. For example, a 23% explicit share in the 2020s means more against the 4.4% overall average.
- They **react to the filters**, so selecting a decade instantly shows how that era compares with the whole catalogue.
- They are placed first in the reading order, so the viewer knows the scale before interpreting trends.

> **Note on Total Artists**
> The 98,504 figure counts distinct artist IDs found in the track-artist table. The cleaned artists table holds 81,776 rows (artists with profile data), so the two numbers differ. Both are correct for what they count.

6. Filters and Why They Are Needed

Three filters on the right-hand panel apply to **every KPI and chart at once**. Each is a single-value dropdown applied to all worksheets.

Filter	Options	Why it is needed
Decade	1920s ... 2020s	Lets the viewer compare eras; all KPIs and charts update for the chosen decade
Explicit	Yes / No	Shows whether explicit songs differ in popularity and style
Content Type	Music / Spoken Word	Removes audiobooks and podcasts so song averages are not distorted

Without filters the dashboard would show one fixed view of 100 years of data. Filters turn it into an **exploration tool**: the same five charts can answer questions about any decade or type of content.

Worked example: Decade = 2020s

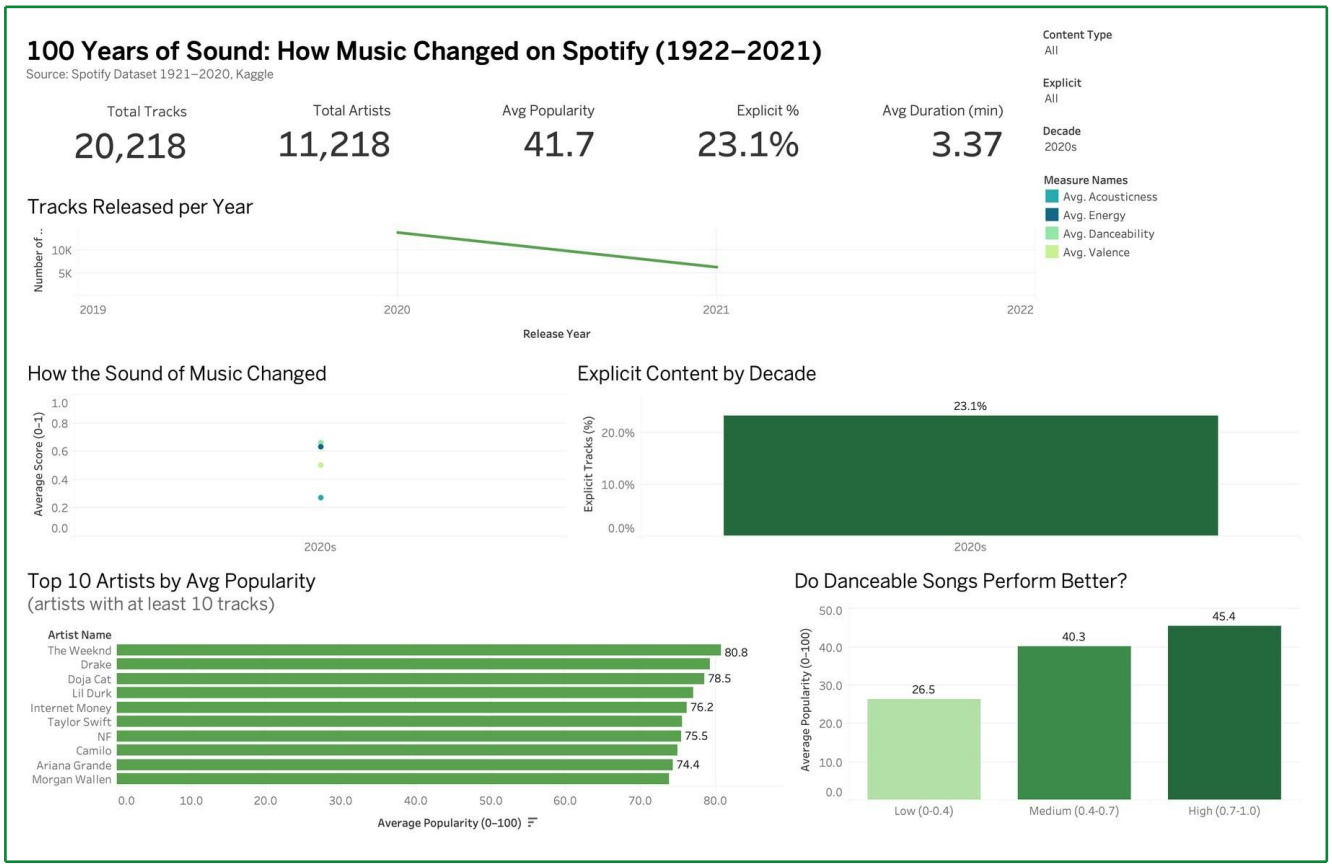
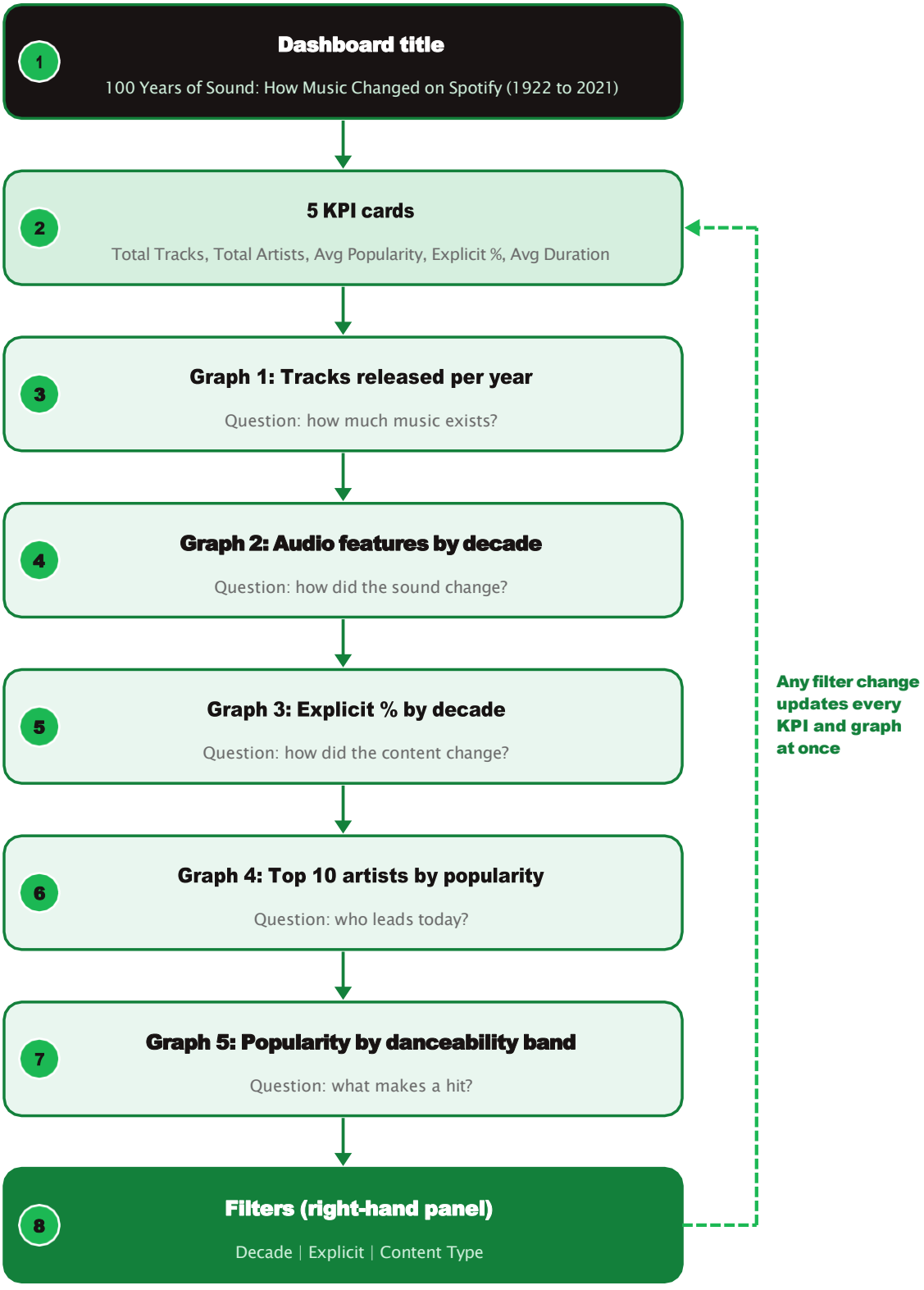


Figure 2: Dashboard with the Decade filter set to 2020s

Measure	All data	Decade = 2020s
Total Tracks	586,600	20,218
Explicit %	4.4%	23.1%
Avg Popularity	27.6	41.7
Top 10 artists	Harry Styles leads (77.8)	Re-ranked: The Weeknd leads (80.8), then Drake, Doja Cat

7. Flowchart: Navigation of the Dashboard

The dashboard is read top to bottom: **Title, KPIs, Graph 1 to Graph 5, Filters**. The graphs tell one story in sequence. Any filter change flows back up (dashed line) and updates every KPI and graph at once.



8. Visualization Interpretations

Each chart is explained in the standard 5-point format: **What do you see, Positive scenario, Negative scenario, Conclusion, Suggestion.**

8.1 Graph 1: Tracks released per year

Chart type: line chart

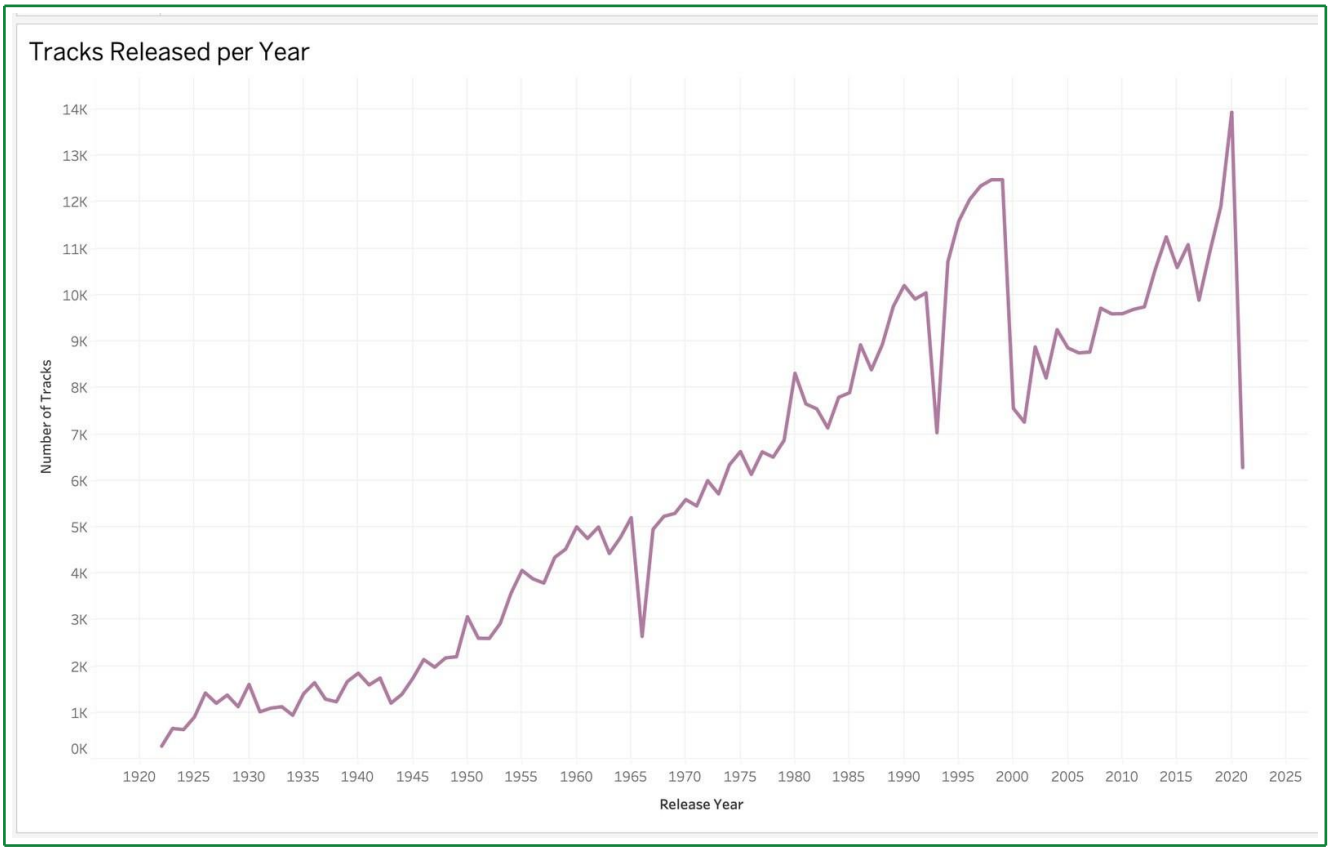


Figure 3: Graph 1 as built in Tableau

1 What do you see?	Releases grow from 276 tracks in 1922 to a peak of 13,937 in 2020. Growth is slow until 1950, then rises steadily, with visible dips in 1966, 1993 and 2000. The final point (2021) is a partial year.
2 Positive scenario	The catalogue keeps expanding, so listeners have more choice than ever and new artists can reach a global audience.
3 Negative scenario	More releases means more competition; each new song has a harder time getting noticed. The dips are probably artefacts of how the dataset was sampled rather than real falls in music production (this was not checked against an outside source).
4 Conclusion	Music production has grown about 50 times over the century (276 to 13,937 a year), with the streaming era (2010s to 2020) producing the most tracks.
5 Suggestion	Artists should plan promotion and playlist strategy, not just release volume, to stand out. Read 2021 as a partial year, and treat single-year dips with caution.

8.2 Graph 2: Audio features by decade

Chart type: multi-line chart

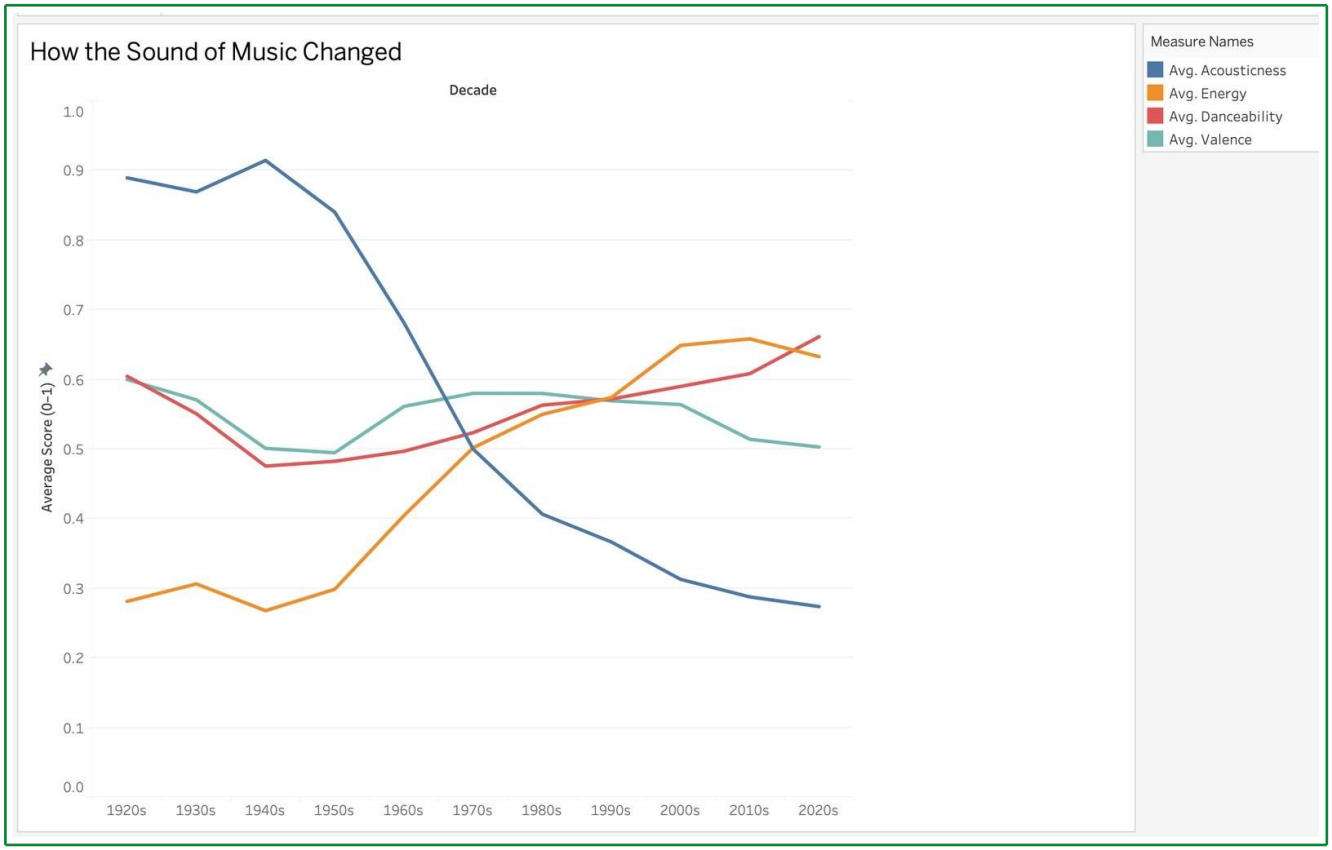


Figure 4: Graph 2 as built in Tableau

1 What do you see?	Acousticness falls from 0.89 (1920s) to 0.27 (2020s), while energy rises from 0.28 to 0.63 and danceability from 0.48 (1940s) to 0.66. The acousticness and energy lines cross in the 1970s. Valence peaks around 0.58 (1970s and 1980s), then drops to 0.50.
2 Positive scenario	Modern music is more energetic and danceable, which suits streaming, fitness and party playlists.
3 Negative scenario	Falling valence shows music has become less cheerful in mood, and acoustic, instrumental styles are being pushed aside.
4 Conclusion	Music has shifted from acoustic and calm to electronic, energetic and danceable, with the turning point in the 1970s.
5 Suggestion	Producers targeting mainstream audiences should favour high-energy, danceable sound; niche acoustic genres need targeted playlists to reach listeners.

8.3 Graph 3: Explicit % by decade

Chart type: bar chart

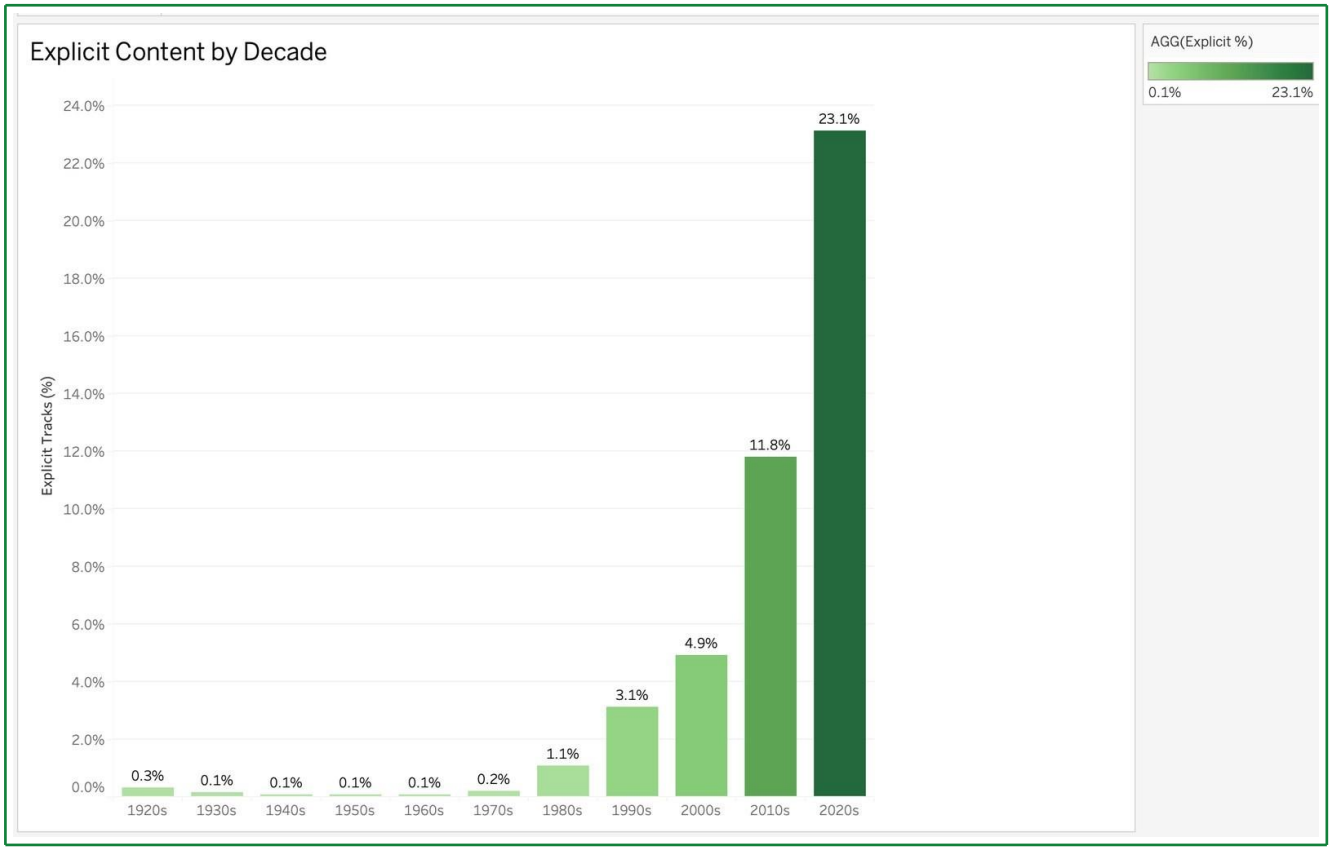


Figure 5: Graph 3 as built in Tableau

1 What do you see?	Explicit tracks stay at or below about 1% until the 1980s, then rise to 3.1% (1990s), 4.9% (2000s), 11.8% (2010s) and 23.1% (2020s).
2 Positive scenario	Artists now have more creative freedom to express real experiences, especially in hip-hop and rap.
3 Negative scenario	Nearly 1 in 4 new songs is explicit, which matters for younger listeners and family-friendly platforms.
4 Conclusion	The explicit share has risen about seven-fold since the 1990s (3.1% to 23.1%): it more than doubled in the 2010s and roughly doubled again in the 2020s. Explicit content is now a mainstream feature of popular music.
5 Suggestion	Streaming platforms should keep strong parental controls and clean versions; brands should check explicit flags before using songs in campaigns.

8.4 Graph 4: Top 10 artists by average popularity

Chart type: horizontal bar chart (artists with at least 10 tracks)

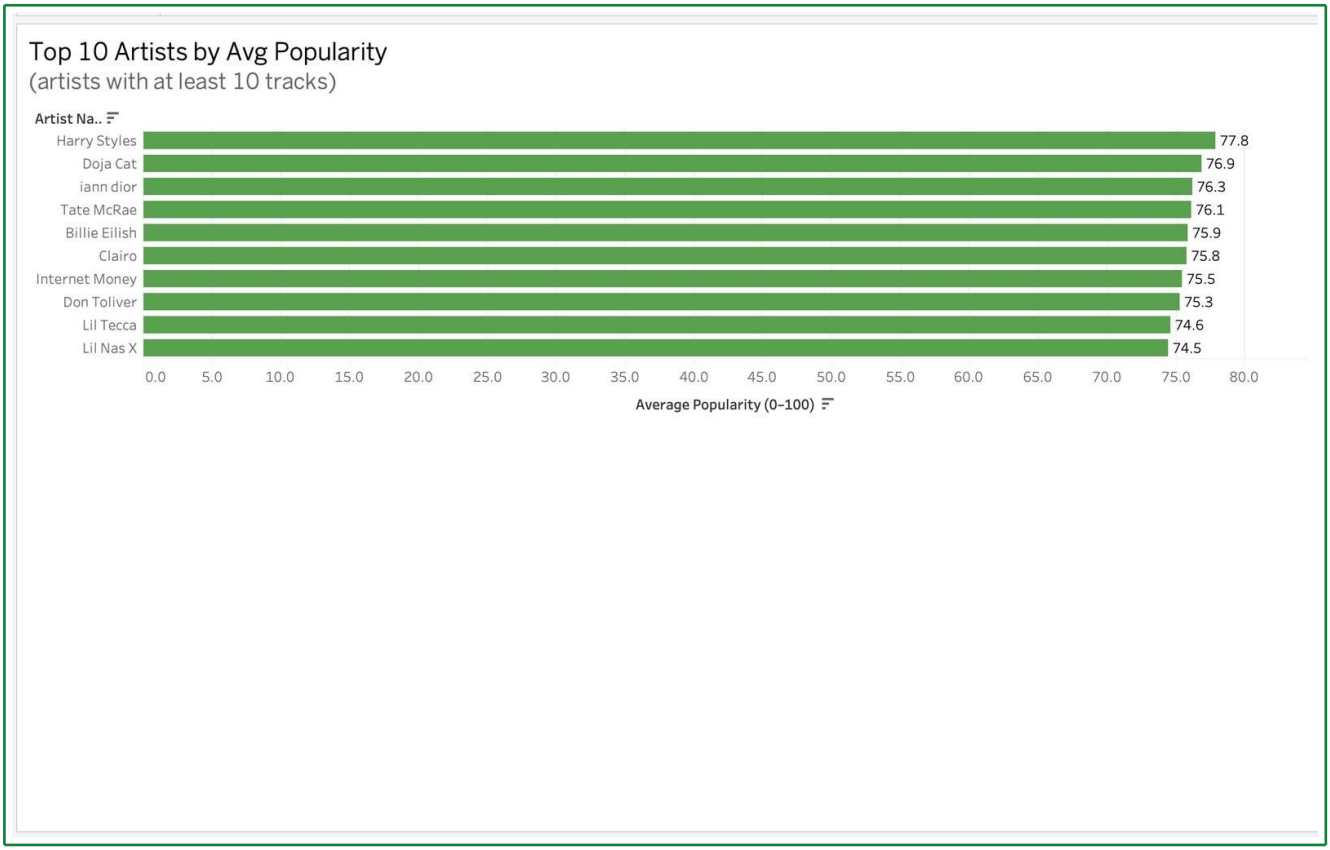


Figure 6: Graph 4 as built in Tableau

1 What do you see?	Harry Styles leads with an average popularity of 77.8, followed by Doja Cat (76.9), iann dior (76.3), Tate McRae (76.1) and Billie Eilish (75.9). All ten are within about 3 points of each other. Only artists with at least 10 tracks are included.
2 Positive scenario	Young and recently emerged artists can reach the top, showing streaming gives new talent a fair chance.
3 Negative scenario	The list is entirely recent pop and hip-hop artists, because Spotify popularity favours recent plays; classic artists are under-represented.
4 Conclusion	Current popularity on Spotify is dominated by young pop and hip-hop artists with consistently strong catalogues.
5 Suggestion	Labels should study these artists' release and collaboration patterns; analysts should remember popularity reflects recent streams, not all-time success.

8.5 Graph 5: Popularity by danceability band

Chart type: bar chart

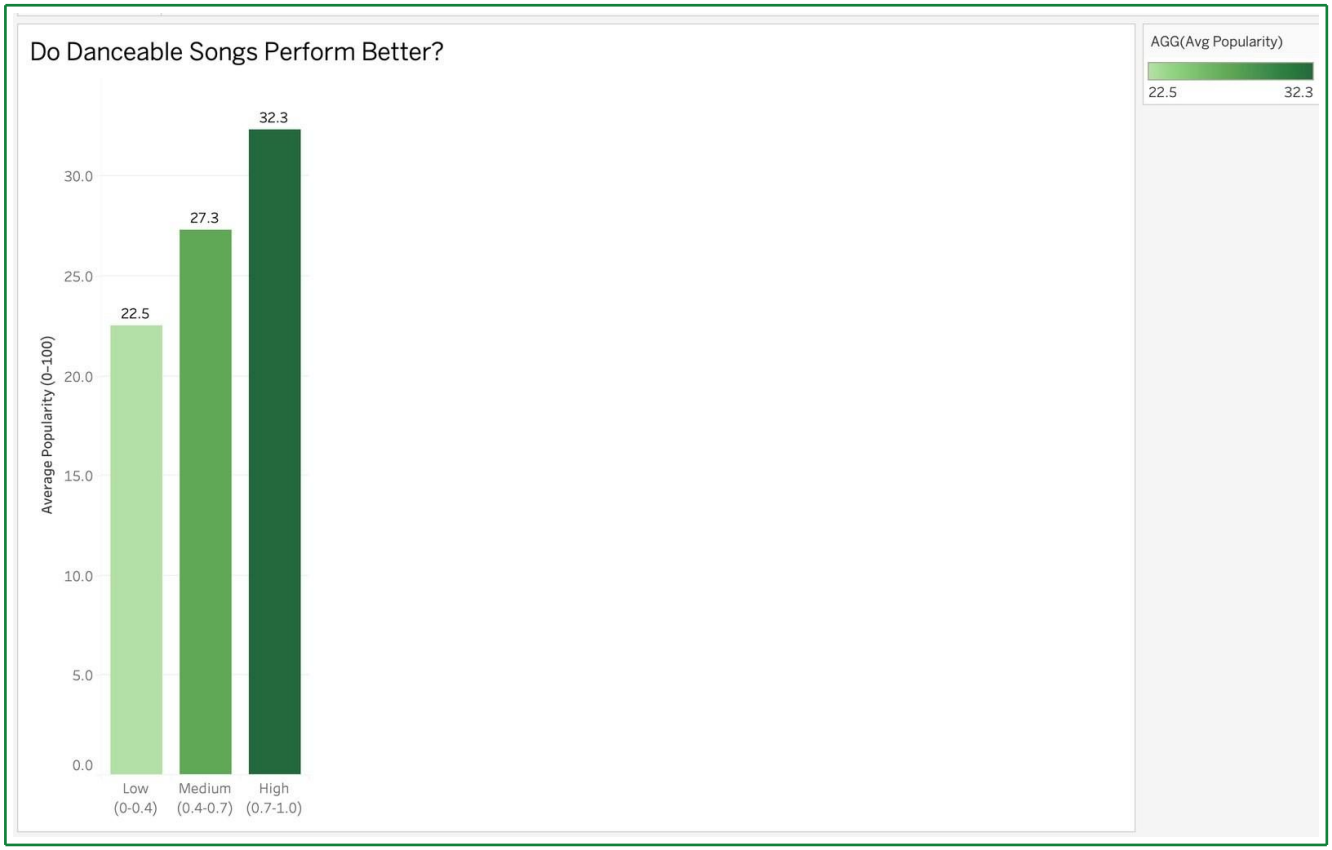


Figure 7: Graph 5 as built in Tableau

1 What do you see?	Average popularity rises with danceability: 22.5 for Low (0 to 0.4), 27.3 for Medium (0.4 to 0.7) and 32.3 for High (0.7 to 1.0) danceability songs.
2 Positive scenario	Artists have a clear, measurable lever: making songs more danceable is linked to higher popularity.
3 Negative scenario	If everyone chases danceability, music may become more uniform. The link is correlation, not proof that danceability causes popularity, and part of the gap may be an era effect: newer songs are both more danceable (Graph 2) and favoured by the popularity score.
4 Conclusion	Highly danceable songs are on average about 44% more popular than low-danceability songs (32.3 vs 22.5).
5 Suggestion	Artists aiming for wide reach should consider danceable rhythms, combined with other features such as energy and modern production. To test the era effect, repeat this chart with the Decade filter set to a single decade.

9. Overall Conclusion and Limitations

Over 100 years, music on Spotify has become **more plentiful, more energetic, more danceable and more explicit**, and the most popular songs today share these traits.

9.1 The story in five links

Graph	Finding
1	Production exploded after 1950 (276 tracks in 1922 to 13,937 in 2020).
2	The sound moved from acoustic to electronic around the 1970s.
3	Explicit content became mainstream after 2010 (11.8% in the 2010s, 23.1% in the 2020s).
4	Young pop and hip-hop artists dominate current popularity.
5	Danceability is linked to higher popularity (22.5 to 32.3).

The filters let viewers test each finding for any decade or content type.

9.2 Limitations to keep in mind

- **Correlation is not causation.** Graph 5 shows an association between danceability and popularity only.
- **Popularity is recency-weighted.** It reflects recent plays, so Graph 4 and Graph 5 favour recent music.
- **2021 is a partial year** (data ends 16 April 2021), and 2020s figures cover only about a year and a half.
- **Releases, not unique songs.** About 60,000 re-releases and remasters are kept as separate tracks.
- **Sampling gaps.** The dips in Graph 1 are likely dataset artefacts and were not verified against another source.
- **Averages hide spread.** Mean popularity and mean audio features do not show how much individual songs vary.

9.3 Possible future work

- Add a genre view using the cleaned genre column (74% of artists are currently Unknown, so coverage must improve first).
- Add a scatter of energy against danceability coloured by popularity to separate the two effects.
- Control for era by comparing popularity by danceability band within each decade.

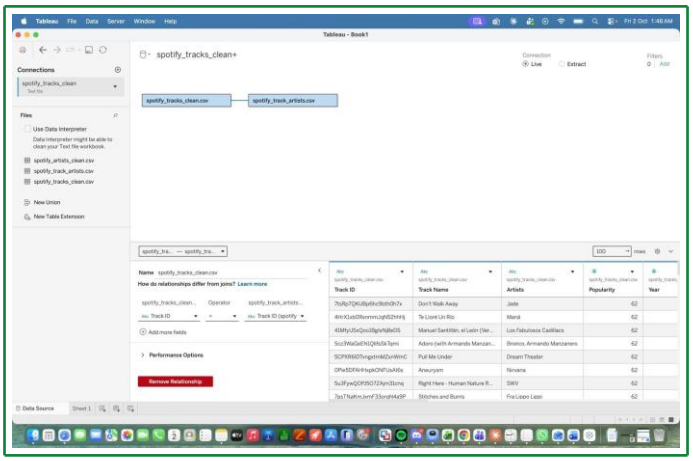
Appendix: Build Procedure in Tableau

The dashboard was built in five phases. The file was saved as **Spotify_Dashboard.twbx**.

Phase 1: Connect the data

- Open Tableau, then Connect, To a File, Text file, and select spotify_tracks_clean.csv.
- Drag spotify_track_artists.csv onto the canvas beside it.
- Click the relationship line and set Track ID = Track ID.
- Check Data Interpreter is off and the field types are correct (Year as number, Decade as text).

Result: both tables are related on Track ID = Track ID using a live connection. Year and Popularity read as numbers; Track ID, Track Name and Artists read as text.

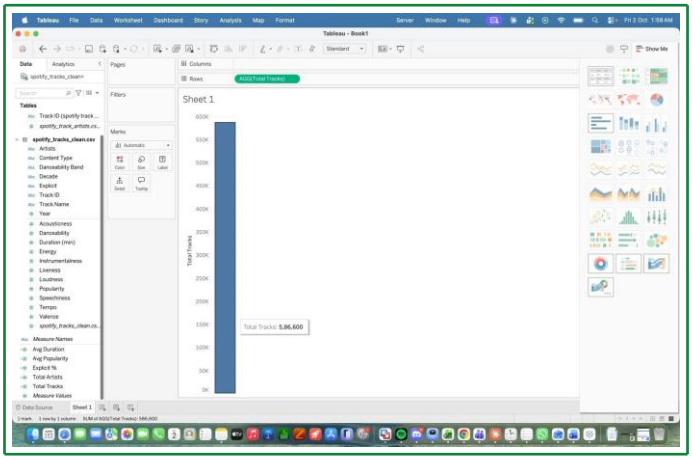


Checkpoint 1: Data Source page with both tables related

Phase 2: Create calculated fields

- Analysis, Create Calculated Field for each KPI: Total Tracks, Total Artists, Avg Popularity, Explicit %, Avg Duration (formulas in Section 5).
- Set Explicit % number format to Percentage (1 decimal).
- Confirm Year is a dimension.

Result: all five calculated fields appear in the Data pane. Total Tracks returns 586,600, matching the cleaned dataset.

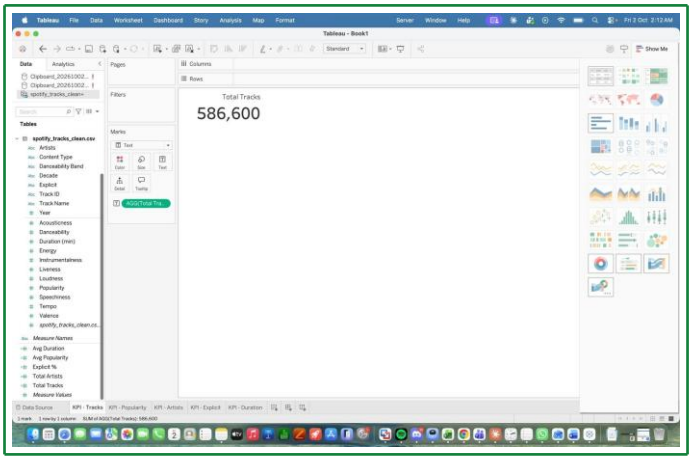


Checkpoint 2: Data pane listing the five new fields

Phase 3: Build the KPI sheets

- Open a new sheet per KPI and drag the measure onto Text on the Marks card.
- Format the number at 24 pt bold and add a small title (e.g. Total Tracks).
- Rename each sheet (KPI – Tracks, KPI – Artists, and so on).

Result: the Total Tracks card shows 586,600 using a Text mark with a two-line label. The workbook locale was set to English (United States) so numbers use standard thousands separators. The other four KPI sheets were created by duplicating this sheet.



Checkpoint 3: a finished KPI sheet

Phase 4: Build the five charts

Graph	Columns	Rows	Mark / setting
1	Year (continuous)	Total Tracks	Line
2	Decade	Measure Values (AVG of Energy, Danceability, Acousticness, Valence)	Measure Names on Color; line
3	Decade	Explicit %	Bar with labels
4	AVG(Popularity)	Artist Name	Filter Total Tracks at least 10; Top 10 by AVG(Popularity); sort descending; labels
5	Danceability Band	AVG(Popularity)	Bar with labels; manual sort Low, Medium, High

Result: all five charts were built and checked against the expected values (screenshots in Section 8). Graph 4 uses a context filter (Total Tracks at least 10) so the Top 10 is chosen only from artists with enough tracks; Graph 5 uses a manual sort.

Phase 5: Assemble the dashboard and filters

- Dashboard, New Dashboard, Size Fixed 1400 x 900.
- Add a Text object for the title, a horizontal container with the five KPI sheets, then the five charts in two rows.
- Show filters for Decade, Explicit and Content Type; set each to Single Value (dropdown) and apply to all worksheets using related data sources.
- Place the filters in a vertical container on the right.
- Test each filter and check every KPI and chart updates.

Result: the dashboard was exported with Dashboard, Export Image, once with all filters set to All and once with Decade = 2020s (Figures 1 and 2). Every chart is set to Fit, Entire View so nothing scrolls, and the three shared filters were added to context so Graph 4 re-ranks correctly for each decade.